

Ayush Singh

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PROFESSIONAL SUMMARY

Java Backend Developer with 5+ years of experience designing and delivering high-availability, distributed microservices systems at scale. Expert in Spring Boot, Kafka, and cloud-native architecture, with a proven track record of driving measurable business impact — including a 35% reduction in API response times and a 15% reduction in fraudulent transactions. Experienced in Agile/Scrum delivery, cross-functional collaboration, CI/CD pipeline ownership, and mentoring junior engineers. Passionate about system design, observability, and building resilient, low-latency platforms for product-based companies.

EXPERIENCE

Entain Software Development Services (India) Pvt. Ltd. | Senior Software Developer *Nov 2021 – Present*

Project 1: BetMGM Online Sportsbook | Java 8/21 · Spring Boot · Microservices · MySQL · Kafka · Docker · Kubernetes · Datadog

- Architected a high-availability RESTful API layer handling real-time wager placement and live sports data sync across 6+ microservices, reducing average API response time by 35% through Redis caching and query optimization.
- Led end-to-end Agile/Scrum delivery for 4 quarterly feature releases — from low-level design to production deployment — coordinating across product, QA, and DevOps teams, achieving on-time delivery in 100% of milestones.
- Drove observability improvements by integrating Datadog APM dashboards and custom alerting rules across 8 distributed services, reducing mean time to detect (MTTD) production incidents from ~40 min to under 10 min.
- Owned MySQL schema design and query optimization for high-throughput betting data; improved critical read query performance by 40% via indexing strategy and connection pool tuning.
- Established CI/CD pipeline standards using Jenkins and Docker with blue-green deployment strategy, cutting release cycle time from 2 hours to under 20 minutes with zero-downtime deployments.
- Collaborated with product owners and DevOps to translate business requirements into technical designs, ensuring seamless third-party API integration for live game stats and real-time odds updates.

Project 2: Automated Fraud Prevention via LexisNexis | Java 8 · Spring Boot · Microservices · SQL · REST APIs

- Designed and owned distributed microservices integrating LexisNexis risk profiling APIs with internal ezecash reporting systems, reducing fraudulent transactions by 15% within Q1 post-deployment.
- Led the full delivery lifecycle — requirements gathering through production rollout — for a fraud detection platform serving Risk and Compliance teams, delivering 2 weeks ahead of schedule with zero critical post-launch defects.
- Engineered fault-tolerant service architecture using circuit breaker patterns and retry mechanisms, maintaining 99.9% service uptime under 3× peak transaction load during high-risk periods.
- Optimized complex SQL reporting queries, cutting fraud report generation time by 50% and enabling real-time decision-making for analysts on flagged high-risk transactions.
- Mentored 2 junior developers on microservices best practices and code review standards, reducing average PR review cycle time by 30%.
- Collaborated with stakeholders and Risk/Fraud teams to gather requirements, align on compliance needs, and drive adoption of improved data profiling algorithms for suspicious activity identification.

SKILLS

Backend & Core: Java (JDK 8/21), Spring, Spring Boot, Microservices, REST APIs, Hibernate, JPA

Databases & Caching: MySQL, SQL, MongoDB, Redis

Messaging & Streaming: Kafka

Cloud & Infrastructure: Docker, Kubernetes, Jenkins (CI/CD), Maven, Gradle, Git, Tomcat
Monitoring & Observability: Datadog (APM, Logs, Metrics, Dashboards, Alerting), ELK Stack
System Design: Distributed Systems, High-Availability Architecture, Fault Tolerance, Scalable API Design, Resiliency
AI & Developer Tools: GitHub Copilot, Amazon Q, Kiro (AI-assisted coding, debugging, test generation)
Testing & Quality: JUnit, Mockito, Unit Testing, Integration Testing, API Testing
Tools & Process: Postman, Jira, Agile / Scrum

EDUCATION

Lovely Professional University, Phagwara, Punjab B.Tech — Computer Science & Engineering	2018 – 2021
Lovely Professional University, Phagwara, Punjab B.Tech Diploma — Computer Science & Engineering	2016 – 2018

CERTIFICATIONS & TRAINING

- Summer Internship in C, C++, HTML & Java — Makryto Innovation Pvt. Ltd. (2020)
- Python Programming Training — Multisoft (2019)
- Machine Learning Training — Multisoft (2019)

LANGUAGES

Hindi (Native) | English (Professional Working Proficiency)